PES1UG24AM139

STUDENT CLUB EVENT TICKETING AND BUDGET PORTAL

USE CASE 3: Generate QR Tickets

Actor: System (automatic), Campus Admin (manual override)

Priority: High

Related FRs: FR-003

Preconditions

Event proposal has "Approved - Budget Allocated" status

Event date is confirmed and locked

Attendee registration/ticketing window is open

System has access to QR code generation library

Security encryption keys are active

Postconditions

Unique QR codes generated for each registered attendee

Ticket database records created with:

Event ID, Attendee ID, Ticket ID, QR code, timestamp, venue

Tickets marked as "Active" and ready for download

Email notifications sent to attendees with ticket links

Campus Admin can view ticket generation report

Main Success Scenario

Event date approaches (10 days before event)

System automatically triggers ticket generation process

System queries registered attendees for event (e.g., 150 attendees)

For each attendee, system performs:

Generate unique Ticket ID (e.g., TKT-PROP-2026-1024-001)

Generate QR code containing: [Event ID | Attendee ID | Timestamp | Venue Code]

Encrypt QR code with AES-256 encryption

Store in Ticket DB: ticket_id, attendee_id, qr_hash, status=ACTIVE

System creates ticket file (PDF with QR code + event details)

Ticket files uploaded to secure CDN

For each attendee:

System generates secure download link (valid 24 hours)

Sends email: "Your ticket for [Event Name] is ready. Download: [link]"

Subject: "Your Event Ticket - [Event Date]"

Attendee ID is appended to link for tracking

System logs all ticket generation: "150 tickets generated for PROP-2026-1024"

Campus Admin receives summary report:

Total tickets generated: 150

QR code validity window: [start date] to [event date + 1 day]

Failed generations (if any): 0

Alternate Flow

Alternate Flow A: Attendee Registration Exceeds Capacity

At Step 3, if registrations > venue capacity:

System caps ticket generation at venue capacity (e.g., 150 max)

Remaining 20 registered attendees marked as "Waitlist"

Email sent to waitlist attendees: "You are #1 on waitlist. Standby notification will be sent if space becomes available"

Proceed from Step 4 (generate for 150 only)

Alternate Flow B: Manual Ticket Generation by Campus Admin

Admin can manually trigger ticket generation before 10-day automatic window:

Admin navigates to event and clicks "Generate Tickets Now"

System prompts: "Generate tickets for X attendees. Confirm?"

Admin confirms

Proceed from Step 3 onward

This allows early distribution if needed

Alternate Flow C: Ticket Regeneration (Lost/Corrupted)

If attendee loses ticket or QR code is corrupted:

Attendee clicks "Resend Ticket" in portal

System generates new QR code with same Ticket ID (old code invalidated)

New email sent with updated download link

Old ticket marked as "Revoked"

New ticket marked as "Active"

Log entry: "Ticket TKT-... regenerated for attendee on [date]"

Alternate Flow D: Event Cancelled

If event is cancelled before ticket distribution:

System marks all generated tickets as "Cancelled"

Email sent to attendees: "Event [name] cancelled. Refund initiated."

QR codes invalidated in venue scanner database

End Use Case